[INCH-POUND]
A-A-2931
April 1, 1996
SUPERSEDING
GGG-R-180
January 21, 1992

COMMERCIAL ITEM DESCRIPTION

REAMER, HAND, BROWN AND SHARPE TAPER SOCKET

The General Services Administration has authorized the use of this Commercial Item Description for procurement of the Type XIV, Classes 1 and 2 reamers specified in GGG-R-180 for all federal agencies.

1. SCOPE

This Commercial Item Description covers reamers used in reaming Brown & Sharpe taper sockets.

2. CLASSIFICATION

Class 1 - Straight Fluted.
Class 2 - Spiral Fluted.

3. SALIENT CHARACTERISTICS

3.1. Class 1.

3.1.1 Design. The hand reamer shall have straight flutes, straight round shank with squared end, and shall be right-hand and ground. The reamer shall be similar to figure 1 (A), and conform to the dimensions of table 1 and the applicable tolerances of table 2 (see 7.2).

3.2 Class 2.

3.2.1 Design. The hand reamer shall have spiral flutes, straight round shank with squared end, and shall be left spiral and right-hand cut. The reamer shall be similar to figure 1 (B), and conform to the dimensions of table 1 and the applicable tolerances of table 2 (see 7.2).

3.3 Material. Unless otherwise specified (see 7.2), the reamer shall be made of Molybdenum base types M1, M2, M7 or M10 high speed steel.

3.4 Construction. The cutting section shall taper as specified in table 1. The cutting edges shall be properly relieved and the flutes shall provide for chip clearance. Flute spacing shall be in accordance with the manufacturer's standard practice. The shank shall be integral or joined to the reamer by welding, and shall be concentric with the cutting edges within 0.001 inch total indicator reading (tir) as measured between centers. The chamfer on the cutting end of the reamer is optional.

Beneficial comments, recommendations, additions, clarification, etc. and any data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5110

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3.5 Hardness. The full length of the cutting edges of the reamer shall be hardened and tempered to a Rockwell hardness value from C61 to C67. When required to be tested, the reamer shall be tested for hardness in accordance with ASTM E18.

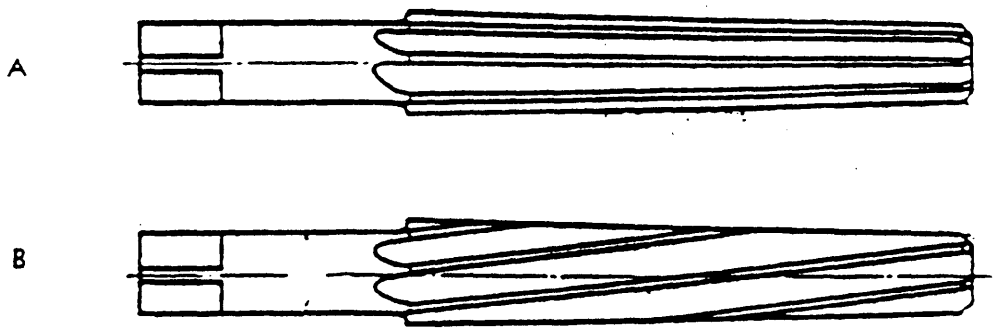


Figure 1. Brown & Sharpe Taper Hand Reamer

Taper No.	Small End Dia.	Large End Dia.	Overall Length	Flute Length	Square Length	Shank Dia.	Size of Square	Taper Per Foot	No. of Flutes
1	0.1974	0.3176	4 3/4	2 7/8	1/4	9/32	0.210	0.50200	4 to 6 incl.
2	0.2474	0.3781	5 1/8	3 1/8	5/16	11/32	0.255	0.50200	4 to 6 incl.
3	0.3099	0.4510	5 1/2	3 3/8	3/8	13/32	0.305	0.50200	4 to 6 incl.
4	0.3474	0.5017	5 7/8	3 11/16	7/16	7/16	0.330	0.50240	4 to 6 incl.
5	0.4474	0.6145	6 3/8	4	1/2	9/16	0.420	0.50160	4 to 6 incl.
6	0.4974	0.6808	6 7/8	4 3/8	5/8	5/8	0.470	0.50329	4 to 6 incl.
7	0.5974	0.8011	7 1/2	4 7/8	3/4	3/4	0.560	0.50147	6 to 8 incl.
8	0.7474	0.9770	8 1/8	5 1/2	13/16	13/16	0.610	0.50100	6 to 8 incl.
9	0.8974	1.1530	8 7/8	6 1/8	7/8	1	0.750	0.50085	6 to 8 incl.
10	1.0420	1.3376	9 3/4	6 7/8	1	1 1/8	0.845	0.51612	6 to 8 incl.

Table 1.

Element	Range Size Number	Tolerances
Length Overall (A)	1 to 7 incl. 8 to 10 incl.	+/- 1/16 +/- 3/32
Length of Flute (B)	1 to 7 incl. 8 to 10 incl.	+/- 1/16 +/- 3/32
Length of Square (C)	1 to 9 incl. 10	+/- 1/32 +/- 1/16
Diameter of Shank (D)	1 to 10 incl.	- 0.0005 to 0.002
Size of Square	1 to 3 incl. 4 to 9 incl. 10	- 0.004 - 0.006 -.0.008

Table 2.

3.6 Performance. The reamer shall cut freely and produce smooth and accurate holes when subjected to hand reaming a minimum of 25 holes in a steel plate of at least 1/2 inch thickness. The testing steel shall conform to FED-STD-66, AISI Number 4140 or 4150, and shall have a Brinell hardness value from 235 to 285. The holes shall be prepared with a taper approximating that of the reamer taper.

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3.7 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g. sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety or appearance.

3.8 Identification marking. The reamer shall be marked with the manufacture's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item. The reamer shall be marked with the words "High Speed" or the abbreviation "HS".

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed
the tolerances specified in the inch-pound units, a request should be made to the Contracting Officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall provide certification and maintain substantiating evidence that all items being offered for acceptance meet the requirements of this Commercial Item Description and conform to the manufacturer's drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

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6. PACKAGING

Requirements for preservation, packing and marking shall be as specified in the contract or order.

7. NOTES

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of reference documents.

Federal Specifications: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.

Federal Acquisition Regulations (FAR): Government Printing Office Superintendent of Documents, Washington, DC 20401-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 43201-4606.

ASME B94.2: the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

ASTM E18: American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and shall include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) Class and taper number required.
(c) Type of material required if other than specified.
(d) When applicable, the appropriate NSN.
(e) If bid samples are required and the number of samples required.
(f) If first article samples are required and the number of samples required.
(g) If lot sampling inspections are required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Examinations. Each sample tool shall be examined for all salient characteristics. The Inspection Level shall be S-3 with an Acceptable quality level (AQL) of 2.5 percent defective.

Testing. Each sample tool shall be tested for all salient characteristics. The Inspection Level for the hardness test shall be S-2 with an Acceptable quality level (AQL) of 2.5 percent defective. The Inspection Level for the performance test shall be S-2 with an Acceptable quality level (AQL) of 4.0 percent defective.

(h) What preservation, packaging, and marking are required.

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7.3 National Stock Numbers (NSNs). NSNs shown may not be indicative of all possible NSNs associated with this CID:

Class 1:

Reamer Size	NSN: 5110-00-	Reamer Size	NSN: 5110-00-
1	186-4203	6	186-4208
2		7	
3	186-4205	8	186-4210
4	186-4206	9	186-4211
5	186-4207	10	186-4212

Class 2:

Reamer Size	NSN: 5110-00-	Reamer Size	NSN: 5110-00-
1		6	
2	186-4204	7	186-4209
3		8	
4		9	
5		10	

Military Interests:
None: DoD has no registered interest in revisions and notices to this Commercial Item Description until further notice.

Preparing Activity
GSA - FSS

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